

Assignment 1: Sexism Detection with Stacked RNNs and Transformers

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Abstract

This report addresses the EXIST 2023(Plaza et al.) Task 2 challenge on sexism detection, focusing on categorizing the intention behind sexist messages into four classes: *Direct*, *Judgemental*, *Reported*, and *Non-sexist*. We evaluate three distinct architectures: a baseline Bidirectional LSTM, a Stacked Bi-LSTM(Graves and Schmidhuber), and a fine-tuned RoBERTa-base Transformer. To address the significant class imbalance, we employed a weighted loss strategy during training. Experimental results on the test set demonstrate that the RoBERTa model achieves the best performance, reaching an Accuracy of 0.72 and a Macro- F_1 score of 0.43 on the test set, outperforming both LSTM baselines (Macro- F_1 0.36 and 0.37 respectively). Despite the higher accuracy, the moderate F_1 score highlights the persistent challenge of correctly classifying minority classes such as 'Reported', even with advanced pre-trained models.

1 Introduction

Fine-grained sexism detection is a subtle and critical task in Natural Language Processing (NLP), necessitating models capable of discerning not just the presence of toxicity, but the specific pragmatic intention behind it. This study addresses the **EXIST 2023 Task 2** challenge, a multi-class classification problem aiming to categorize sexist content into four distinct labels: 'Direct', 'Judgemental', 'Reported', or 'Non-sexist'. Historically, Recurrent Neural Networks (RNNs) have served as the standard baselines for such sequence modeling tasks. However, the field has recently shifted towards Transformer-based architectures, which promise superior handling of long-range dependencies and semantic ambiguity. In this report, we present a comparative analysis between a traditional Bidirectional LSTM pipeline and a Transfer Learning approach based on the **RoBERTa** architecture. Our primary objective is to assess whether the deep,

contextualized representations offered by Transformers can effectively overcome the limitations of recurrent models, particularly when dealing with the severe class imbalance inherent in the "small" version of the provided dataset.

2 System Description

We designed and implemented a set of three distinct architectures to isolate the impact of model complexity on classification performance.

Architectural Framework The first two models represent our recurrent baselines. Both rely on the GloVe vocabulary, which contains approximately 400,000 token embeddings, augmented with additional tokens discovered during tokenization of the training set. A critical detail in our embedding strategy concerns the handling of Out-Of-Vocabulary (OOV) terms: specifically, OOV token embeddings are initialized randomly. This ensures that unknown words possess a learnable vector representation during training rather than being discarded or mapped to a static zero vector.

- **Bi-LSTM (Baseline):** The first model consists of a single Bidirectional Long Short-Term Memory (LSTM) layer, which processes the input sequence in both forward and backward directions to capture temporal context.
- **Stacked Bi-LSTM:** To test the hypothesis that deeper networks can extract more abstract features, we extended the baseline by stacking a second Bidirectional LSTM layer on top of the first.

Conversely, our third system leverages the RoBERTa-base Transformer (Liu et al.). Unlike the LSTM models which learn embeddings from scratch, RoBERTa utilizes a pre-trained tokenizer and context-aware embeddings, fine-tuned here for sequence classification using the Hugging Face library.

3 Experimental Setup and Results

All experiments were conducted on the "small" version of the EXIST dataset (using only the English language sentences), using the AdamW optimizer to ensure effective convergence. To guarantee the statistical robustness of our findings, we ran the training process with 5 different random seeds (42, 123, 456, 7, 888) and averaged the results. Regarding the training duration, we did not employ early stopping; instead, all models were trained for a fixed duration of 20 epochs, and the best-performing checkpoint was selected based on validation metrics. The hyperparameters were tuned as follows:

- **Recurrent Models:** We used a batch size of 256 and a fixed sequence length of 62 tokens.
- **Transformer:** RoBERTa was fine-tuned with a smaller batch size of 32, a learning rate of $2e^{-5}$, and a weight decay of 0.01.

Model	Accuracy	Macro-F1
Bi-LSTM (Baseline)	0.60	0.36
Stacked Bi-LSTM	0.64	0.37
RoBERTa-base	0.72	0.43

Table 1: Comparative results on the Test Set. RoBERTa outperforms recurrent baselines in accuracy, though the F1 score remains limited.

Table 1 summarizes the performance on the Test set. The results demonstrate that RoBERTa-base significantly outperforms the recurrent architectures, achieving the highest Accuracy (0.72). However, the improvement in Macro-F1 score on test set is marginal (0.43), indicating a persistent challenge in correctly retrieving minority classes despite the architectural improvements.

4 Discussion

Quantitative Analysis Our analysis confirms that pre-trained Transformers are superior to RNNs for this task, yielding a +12% accuracy boost over the Stacked LSTM. However, a critical observation is the discrepancy between the high Accuracy and the stalled Macro-F1 score. All models exhibit a bias towards the majority class. One factor influencing the baseline performance is the vocabulary coverage. Our analysis of the GloVe tokenizer usage reveals a non-negligible impact of unknown terms, resulting in a total of 6.91% unknown tokens in the test set. This loss of semantic information likely hinders the LSTM’s ability to classify subtle

examples, a problem less severe for RoBERTa’s sub-word tokenizer.

Error Analysis We identified two systematic failure modes through confusion matrix analysis. First, semantic ambiguity: models struggle to distinguish between ‘Direct’ and ‘Judgemental’ sexism. These classes share linguistic markers of aggression, and without broader context, the decision boundary remains blurred. Second, the ‘Reported’ bottleneck: in instances involving quotations (e.g., *"He said..."*), models frequently misclassify the text based on the quoted content rather than the reporting act, failing to attend to the syntactic cues of indirect speech.

5 Conclusion

Our comparative analysis confirms that RoBERTa-base provides a more robust foundation for sexism detection than recurrent baselines. However, the persistent gap between Accuracy and Macro-F1 reveals that architectural sophistication is not a silver bullet for data scarcity. The ‘Reported’ class remains largely intractable with the current support. Future work should therefore prioritize data-centric solutions, such as data augmentation or domain-adaptive pre-training (e.g., HateBERT), to better align the model’s representation with the nuances of the minority classes.

In addition, improved preprocessing strategies could benefit models relying on word-based tokenizers, such as the GloVe-based approach used in this work. Several unknown tokens were found to consist of otherwise valid vocabulary words concatenated with non-alphanumeric characters. Normalizing or separating punctuation during preprocessing could reduce the number of out-of-vocabulary tokens and partially mitigate their negative impact on downstream classification performance.

References

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